

#	Old Table(s) (30)	New Table (13)	Normalization & Design Rationale
1	department	department	Retained & Simplified: Preserved hierarchy (<code>parent_department_id</code>).
2	designation	designation	Retained: Master catalog for job titles and levels.
3	employee, user_account, employment_assignment, employment_statuses_history	employee	Consolidated (3NF): Current department, designation, reporting manager, and login credentials (<code>username</code> , <code>password_hash</code> , <code>role</code>) belong directly to <code>employee</code> (1-to-1). Historical changes are captured by <code>audit_log</code> .
4	role, permission, user_role, role_permission	Consolidated into <code>employee.role</code>	Streamlined: The project defines 3 roles (<code>EMPLOYEE</code> , <code>MANAGER</code> , <code>HR_ADMIN</code>). An ENUM/CHECK constraint on <code>employee</code> eliminates 4 tables without losing authorization control.
5	leave_type, leave_policy, leave_policy_entitlement, employee_leave_policy	leave_type	De-abstracted: The 3 intermediate policy tables created 3 levels of unnecessary indirection. Default entitlements (e.g., Casual Leave = 12/yr) and carry-forward rules live directly in <code>leave_type</code> .
6	leave_balance	leave_balance	Retained (3NF): Tracks opening, accrued, used, and closing balance per (<code>employee_id</code> , <code>leave_type_id</code> , <code>leave_year</code>).
7	leave_application, leave_approval	leave_application	Merged (3NF): Scope requires single-level manager approval. Approver ID, approval status, actioned timestamp, and manager remarks belong directly on the application entity.

8	attendance, attendance_summary	attendance	Normalized (3NF): attendance_summary stored derived transitive totals (present_days, lop_days), risking update anomalies. Monthly aggregation is computed on-demand or during the payroll run.
9	salary_component	salary_component	Retained: Master list of Earnings and Deductions (Basic, HRA, PF, ESI, PT, LOP, etc.).
10	salary_structure, salary_structure_component, employee_salary	employee_salary & employee_salary_detail	Normalized: Replaced 3-tier structure indirection with employee-level salary assignment and itemized component values/percentages.
11	tax_slab	tax_slab	Retained: Progressive income tax slabs per financial year and regime.
12	payroll_rule, payroll_rule_version	payroll_rule	Consolidated: Replaced separate versioning table with a clean rule definition table.
13	payroll_run, employee_payroll, payroll_item	payroll_run, employee_payroll, payroll_item	Retained (Core Engine): The batch run header, individual employee monthly payslip summary, and itemized component breakdown are fully normalized in 3NF.
14	audit_log	audit_log	Retained: Centralized audit trail for all transactional changes.

2. Normalization Analysis (1NF

→

→ 2NF

→

→ 3NF)

1. First Normal Form (1NF):

- All attributes contain atomic values (eliminated free-form JSON strings for core business attributes).
- Each table has a distinct primary key (`BIGINT AUTO_INCREMENT` or composite natural keys where applicable).

2. Second Normal Form (2NF):

- All non-key attributes are fully functionally dependent on the entire primary key (e.g., in `leave_balance`, `closing_balance` depends on (`employee_id`, `leave_type_id`, `leave_year`)). No partial key dependencies.

3. Third Normal Form (3NF):

- **No Transitive Dependencies (**
- $X \rightarrow Y \rightarrow Z$
- $X \rightarrow Y \rightarrow Z$):
 - In the old schema, `attendance_summary` stored `lop_days` and `absent_days`, which were transitively dependent on the individual `attendance` and `leave_application` records. This was removed.
 - `leave_approval` had a transitive dependency back to `employee_id` through `leave_application_id`. Integrating the single-stage manager approval into `leave_application` eliminates this transitive dependency and prevents orphaned approval rows.

4. Boyce-Codd Normal Form (BCNF):

- Every determinant in every relation is a candidate key.

```
-- =====
-- LEAVE AND PAYROLL MANAGEMENT SYSTEM (LPMS)
-- NORMALIZED DATABASE SCHEMA (3NF) — 13 TABLES
-- =====
```

```
CREATE DATABASE IF NOT EXISTS lpms;
USE lpms;
```

```
-- =====
-- MODULE 1 & 2 : EMPLOYEE & AUTHENTICATION
-- =====
```

```
-- 1. DEPARTMENT
```

```

CREATE TABLE department (
    department_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    department_code VARCHAR(30) NOT NULL UNIQUE,
    department_name VARCHAR(100) NOT NULL UNIQUE,
    description VARCHAR(255),
    parent_department_id BIGINT UNSIGNED NULL,
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
    CURRENT_TIMESTAMP,

    CONSTRAINT fk_dept_parent
        FOREIGN KEY (parent_department_id)
        REFERENCES department(department_id)
        ON DELETE SET NULL ON UPDATE CASCADE
);

```

-- 2. DESIGNATION

```

CREATE TABLE designation (
    designation_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    designation_code VARCHAR(30) NOT NULL UNIQUE,
    designation_name VARCHAR(100) NOT NULL UNIQUE,
    job_level VARCHAR(30),
    description VARCHAR(255),
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
    CURRENT_TIMESTAMP
);

```

-- 3. EMPLOYEE (Master Employee + Current Assignment + Auth merged)

```

CREATE TABLE employee (
    employee_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    employee_code VARCHAR(30) NOT NULL UNIQUE,
    first_name VARCHAR(60) NOT NULL,
    last_name VARCHAR(60) NOT NULL,
    gender ENUM('MALE', 'FEMALE', 'OTHER') NOT NULL,
    date_of_birth DATE,
    email VARCHAR(255) NOT NULL UNIQUE,
    phone VARCHAR(20),
    date_of_joining DATE NOT NULL,
    date_of_exit DATE,
    employment_type ENUM('FULL_TIME', 'PART_TIME', 'PROBATION', 'INTERN') NOT NULL
    DEFAULT 'FULL_TIME',

```

```
employment_status ENUM('ACTIVE', 'ON_LEAVE', 'RESIGNED', 'TERMINATED') NOT NULL
DEFAULT 'ACTIVE',
```

```
-- Organizational assignments
```

```
department_id BIGINT UNSIGNED NOT NULL,
designation_id BIGINT UNSIGNED NOT NULL,
reporting_manager_id BIGINT UNSIGNED NULL,
```

```
-- Authentication & RBAC (3 Roles)
```

```
username VARCHAR(100) NOT NULL UNIQUE,
password_hash VARCHAR(255) NOT NULL,
role ENUM('EMPLOYEE', 'MANAGER', 'HR_ADMIN') NOT NULL DEFAULT 'EMPLOYEE',
is_active BOOLEAN NOT NULL DEFAULT TRUE,
last_login_at DATETIME,
```

```
created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
```

```
CONSTRAINT fk_emp_dept FOREIGN KEY (department_id) REFERENCES
department(department_id),
```

```
CONSTRAINT fk_emp_desig FOREIGN KEY (designation_id) REFERENCES
designation(designation_id),
```

```
CONSTRAINT fk_emp_manager FOREIGN KEY (reporting_manager_id) REFERENCES
employee(employee_id) ON DELETE SET NULL,
```

```
CONSTRAINT chk_emp_exit_date CHECK (date_of_exit IS NULL OR date_of_exit >=
date_of_joining)
);
```

```
-- =====
-- MODULE 3 : LEAVE MANAGEMENT
-- =====
```

```
-- 4. LEAVE TYPE
```

```
CREATE TABLE leave_type (
leave_type_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
leave_code VARCHAR(30) NOT NULL UNIQUE,
leave_name VARCHAR(100) NOT NULL,
description VARCHAR(255),
annual_entitlement DECIMAL(5,2) NOT NULL DEFAULT 12.00,
is_paid BOOLEAN NOT NULL DEFAULT TRUE,
allows_half_day BOOLEAN NOT NULL DEFAULT TRUE,
allows_carry_forward BOOLEAN NOT NULL DEFAULT FALSE,
max_carry_forward DECIMAL(5,2) NOT NULL DEFAULT 0.00,
```

```
is_active BOOLEAN NOT NULL DEFAULT TRUE,  
created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP  
);
```

-- 5. LEAVE BALANCE

```
CREATE TABLE leave_balance (  
    leave_balance_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    employee_id BIGINT UNSIGNED NOT NULL,  
    leave_type_id BIGINT UNSIGNED NOT NULL,  
    leave_year SMALLINT UNSIGNED NOT NULL,  
    opening_balance DECIMAL(5,2) NOT NULL DEFAULT 0.00,  
    accrued DECIMAL(5,2) NOT NULL DEFAULT 0.00,  
    used DECIMAL(5,2) NOT NULL DEFAULT 0.00,  
    closing_balance DECIMAL(5,2) GENERATED ALWAYS AS (opening_balance + accrued -  
used) STORED,  
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE  
CURRENT_TIMESTAMP,
```

```
    CONSTRAINT fk_bal_emp FOREIGN KEY (employee_id) REFERENCES  
employee(employee_id) ON DELETE CASCADE,  
    CONSTRAINT fk_bal_type FOREIGN KEY (leave_type_id) REFERENCES  
leave_type(leave_type_id) ON DELETE RESTRICT,  
    CONSTRAINT uq_emp_leave_year UNIQUE (employee_id, leave_type_id, leave_year)  
);
```

-- 6. LEAVE APPLICATION (With integrated Single-Stage Approval)

```
CREATE TABLE leave_application (  
    leave_application_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    employee_id BIGINT UNSIGNED NOT NULL,  
    leave_type_id BIGINT UNSIGNED NOT NULL,  
    start_date DATE NOT NULL,  
    end_date DATE NOT NULL,  
    total_days DECIMAL(5,2) NOT NULL,  
    reason VARCHAR(500) NOT NULL,  
    status ENUM('PENDING', 'APPROVED', 'REJECTED', 'CANCELLED') NOT NULL DEFAULT  
'PENDING',
```

-- Integrated approval workflow attributes

```
approver_id BIGINT UNSIGNED NULL,  
actioned_at DATETIME NULL,  
manager_comments VARCHAR(500) NULL,
```

```
created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
```

```
updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
```

```
CONSTRAINT fk_app_emp FOREIGN KEY (employee_id) REFERENCES
employee(employee_id),
CONSTRAINT fk_app_type FOREIGN KEY (leave_type_id) REFERENCES
leave_type(leave_type_id),
CONSTRAINT fk_app_approver FOREIGN KEY (approver_id) REFERENCES
employee(employee_id),
CONSTRAINT chk_app_dates CHECK (end_date >= start_date),
CONSTRAINT chk_app_days CHECK (total_days > 0)
);
```

```
-- =====
-- MODULE 4 : ATTENDANCE MANAGEMENT
-- =====
```

```
-- 7. ATTENDANCE (Loss of Pay calculation feeds directly from here)
CREATE TABLE attendance (
    attendance_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    employee_id BIGINT UNSIGNED NOT NULL,
    attendance_date DATE NOT NULL,
    check_in DATETIME NULL,
    check_out DATETIME NULL,
    status ENUM('PRESENT', 'ABSENT', 'HALF_DAY', 'ON_LEAVE', 'HOLIDAY', 'WEEKEND')
NOT NULL,
    worked_minutes INT UNSIGNED DEFAULT 0,
    overtime_minutes INT UNSIGNED DEFAULT 0,
    remarks VARCHAR(255) NULL,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT fk_att_emp FOREIGN KEY (employee_id) REFERENCES
employee(employee_id) ON DELETE CASCADE,
    CONSTRAINT uq_att_emp_date UNIQUE (employee_id, attendance_date),
    CONSTRAINT chk_att_checkout CHECK (check_out IS NULL OR check_in IS NULL OR
check_out >= check_in)
);
```

```
-- =====
-- MODULE 5 & 7 : PAYROLL CONFIGURATION & RULE ENGINE
-- =====
```

```
-- 8. SALARY COMPONENT
CREATE TABLE salary_component (
```

```

component_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
component_code VARCHAR(40) NOT NULL UNIQUE,
component_name VARCHAR(100) NOT NULL,
component_type ENUM('EARNING', 'DEDUCTION', 'EMPLOYER_CONTRIBUTION') NOT
NULL,
calculation_method ENUM('FLAT', 'PERCENTAGE', 'FORMULA') NOT NULL,
is_taxable BOOLEAN NOT NULL DEFAULT TRUE,
is_pf_applicable BOOLEAN NOT NULL DEFAULT FALSE,
is_esi_applicable BOOLEAN NOT NULL DEFAULT FALSE,
is_active BOOLEAN NOT NULL DEFAULT TRUE,
created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
);

```

-- 9. EMPLOYEE SALARY (Current Salary Master & Structure)

```

CREATE TABLE employee_salary (
    employee_salary_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    employee_id BIGINT UNSIGNED NOT NULL,
    annual_ctc DECIMAL(15,2) NOT NULL,
    monthly_gross DECIMAL(15,2) NOT NULL,
    effective_from DATE NOT NULL,
    effective_to DATE NULL,
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

```

```

    CONSTRAINT fk_esal_emp FOREIGN KEY (employee_id) REFERENCES
employee(employee_id),
    CONSTRAINT chk_esal_dates CHECK (effective_to IS NULL OR effective_to >=
effective_from)
);

```

-- 10. EMPLOYEE SALARY DETAIL (Itemized Component Breakdown per Employee)

```

CREATE TABLE employee_salary_detail (
    salary_detail_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    employee_salary_id BIGINT UNSIGNED NOT NULL,
    component_id BIGINT UNSIGNED NOT NULL,
    amount DECIMAL(15,2) NOT NULL,

```

```

    CONSTRAINT fk_esald_salary FOREIGN KEY (employee_salary_id) REFERENCES
employee_salary(employee_salary_id) ON DELETE CASCADE,
    CONSTRAINT fk_esald_comp FOREIGN KEY (component_id) REFERENCES
salary_component(component_id),
    CONSTRAINT uq_esal_comp UNIQUE (employee_salary_id, component_id)
);

```

-- 11. TAX SLAB & RULES (Unified Rule Configuration)

```
CREATE TABLE tax_slab (  
    tax_slab_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    tax_regime ENUM('OLD', 'NEW') NOT NULL DEFAULT 'NEW',  
    financial_year VARCHAR(9) NOT NULL, -- e.g., '2025-2026'  
    slab_order INT NOT NULL,  
    lower_limit DECIMAL(15,2) NOT NULL,  
    upper_limit DECIMAL(15,2) NULL, -- NULL indicates no upper cap  
    tax_rate DECIMAL(5,2) NOT NULL, -- e.g. 5.00 for 5%  
    cess_rate DECIMAL(5,2) NOT NULL DEFAULT 4.00,  
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
    CONSTRAINT uq_slab UNIQUE (tax_regime, financial_year, slab_order)  
);
```

```
-- =====  
-- MODULE 5 : PAYROLL PROCESSING & TRANSACTIONS  
-- =====
```

-- 12. PAYROLL RUN

```
CREATE TABLE payroll_run (  
    payroll_run_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    payroll_year SMALLINT UNSIGNED NOT NULL,  
    payroll_month TINYINT UNSIGNED NOT NULL,  
    status ENUM('DRAFT', 'PROCESSED', 'APPROVED', 'PAID') NOT NULL DEFAULT 'DRAFT',  
    processed_by BIGINT UNSIGNED NOT NULL,  
    approved_by BIGINT UNSIGNED NULL,  
    processed_at DATETIME NULL,  
    approved_at DATETIME NULL,  
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  
    CONSTRAINT fk_prun_proc FOREIGN KEY (processed_by) REFERENCES  
employee(employee_id),  
    CONSTRAINT fk_prun_appr FOREIGN KEY (approved_by) REFERENCES  
employee(employee_id),  
    CONSTRAINT uq_prun_period UNIQUE (payroll_year, payroll_month),  
    CONSTRAINT chk_prun_month CHECK (payroll_month BETWEEN 1 AND 12)  
);
```

-- 13. EMPLOYEE PAYROLL (Monthly Payslip Summary)

```
CREATE TABLE employee_payroll (  
    employee_payroll_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
    payroll_run_id BIGINT UNSIGNED NOT NULL,  
    employee_id BIGINT UNSIGNED NOT NULL,
```

```

-- Derived Attendance & LOP parameters for the month
working_days DECIMAL(4,2) NOT NULL,
present_days DECIMAL(4,2) NOT NULL,
paid_leave_days DECIMAL(4,2) NOT NULL DEFAULT 0,
lop_days DECIMAL(4,2) NOT NULL DEFAULT 0,

-- Financial Totals
gross_earnings DECIMAL(15,2) NOT NULL,
lop_deduction DECIMAL(15,2) NOT NULL DEFAULT 0.00,
total_deductions DECIMAL(15,2) NOT NULL,
net_pay DECIMAL(15,2) NOT NULL,

payment_status ENUM('UNPAID', 'PAID') NOT NULL DEFAULT 'UNPAID',
created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

CONSTRAINT fk_epay_run FOREIGN KEY (payroll_run_id) REFERENCES
payroll_run(payroll_run_id) ON DELETE CASCADE,
CONSTRAINT fk_epay_emp FOREIGN KEY (employee_id) REFERENCES
employee(employee_id),
CONSTRAINT uq_run_employee UNIQUE (payroll_run_id, employee_id)
);

-- 14. PAYROLL ITEM (Itemized lines on the payslip: Basic, HRA, PF, Tax, etc.)
CREATE TABLE payroll_item (
    payroll_item_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    employee_payroll_id BIGINT UNSIGNED NOT NULL,
    component_id BIGINT UNSIGNED NOT NULL,
    amount DECIMAL(15,2) NOT NULL,

    CONSTRAINT fk_pitem_epay FOREIGN KEY (employee_payroll_id) REFERENCES
employee_payroll(employee_payroll_id) ON DELETE CASCADE,
    CONSTRAINT fk_pitem_comp FOREIGN KEY (component_id) REFERENCES
salary_component(component_id),
    CONSTRAINT uq_epay_component UNIQUE (employee_payroll_id, component_id)
);

-- =====
-- MODULE 6 : AUDIT & SYSTEM LOG
-- =====

-- 15. AUDIT LOG
CREATE TABLE audit_log (
    audit_id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,

```

```
entity_name VARCHAR(50) NOT NULL,  
entity_id BIGINT UNSIGNED NOT NULL,  
action ENUM('INSERT', 'UPDATE', 'DELETE') NOT NULL,  
performed_by BIGINT UNSIGNED NULL,  
performed_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
old_values JSON NULL,  
new_values JSON NULL,
```

```
CONSTRAINT fk_audit_user FOREIGN KEY (performed_by) REFERENCES  
employee(employee_id) ON DELETE SET NULL  
);
```

-- Optimization Indexes

```
CREATE INDEX idx_emp_dept ON employee(department_id);  
CREATE INDEX idx_emp_status ON employee(employment_status);  
CREATE INDEX idx_att_emp_date ON attendance(employee_id, attendance_date);  
CREATE INDEX idx_lapp_status ON leave_application(status);  
CREATE INDEX idx_epay_run ON employee_payroll(payroll_run_id);
```